

Introduction

The AMBA® (Advanced Microcontroller Bus Architecture) AXI (Advanced eXtensible Interface) to AHB-Lite (Advanced High Performance Bus) Bridge translates AXI4 transactions into AHB-Lite transactions. It functions as a slave on the AXI4 interface and as a master on the AHB-Lite interface. The AXI to AHB-Lite Bridge's main use model is to connect the AHB-Lite slaves with AXI masters.

Features

The Xilinx® AXI to AHB-Lite Bridge is a soft IP core with the following features:

AXI4 Slave Interface:

- AXI interface is based on the AXI4 specification
- Connects as a 32/64-bit slave on 32/64-bit AXI4
- Supports
 - 1:1 (AXI:AHB) synchronous clock ratio
 - Incrementing burst transfers (length 1 to 256)
 - Wrapping burst transfers (length 2, 4, 8, and 16)
 - Fixed burst transfers (of length 1 to 16)
 - Narrow transfers
 - Limited cache encoding and limited protection unit support
 - Address/data phase timeout

AHB-Lite Master Interface:

- Connects as a 32/64-bit master on 32/64-bit AHB-Lite interface
- Supports:
 - Single burst transfers
 - Wrapping burst transfers (length 4, 8, and 16)
 - Incrementing burst transfers (length 4, 8, 16, and undefined burst length)
 - Limited protection control
 - Narrow transfers
- AHB-Lite master does not issue incrementing burst transfers that cross 1 kB address boundaries

LogiCORE IP Facts Table					
Core Specifics					
Supported Device Family ⁽¹⁾	Zynq™-7000 ⁽²⁾ , Artix™-7, Virtex®-7, Kintex™-7, Virtex-6, Spartan®-6 ⁽³⁾				
Supported User Interfaces	AXI4, AHB-Lite				
	Resources ^{(3), (4), (5)}				Frequency
Configuration	LUTs	FFs	DSP Slices	Max. Freq.	Block RAM
	See Table 6, Table 7, Table 8, Table 9, and Table 10				0
Provided with Core					
Documentation	Product Specification				
Design Files	VHDL				
Example Design	Not Provided				
Test Bench	Not Provided				
Constraints File	None				
Simulation Model	None				
Tested Design Tools ⁽⁶⁾					
Design Entry Tools	Vivado™ Design Suite ⁽⁷⁾ Xilinx Platform Studio (XPS)				
Simulation	Mentor Graphics ModelSim				
Synthesis Tools	Xilinx Synthesis Technology (XST) Vivado Synthesis				
Provided by Xilinx @ www.xilinx.com/support					

Notes:

1. For a complete list of supported derivative devices, see the [Embedded Edition Derivative Device Support](#).
2. Supported in ISE Design Suite implementations only.
3. For more information, see Virtex-6 Family Overview [Ref 3].
4. For more information, see Spartan-6 Family Overview [Ref 4].
5. For more information, see 7 Series FPGAs Overview [Ref 5].
6. For the supported versions of the tools, see the [Xilinx Design Tools: Release Notes Guide](#).
7. Supports 7 series devices only.

Functional Description

Overview

The AXI to AHB-Lite Bridge translates AXI4 transactions into AHB-Lite transactions. The bridge functions as a slave on the AXI4 interface and as a master on the AHB-Lite interface.

Figure 1 shows the AXI to AHB-Lite Bridge block diagram. The modules are described in the subsequent sections.

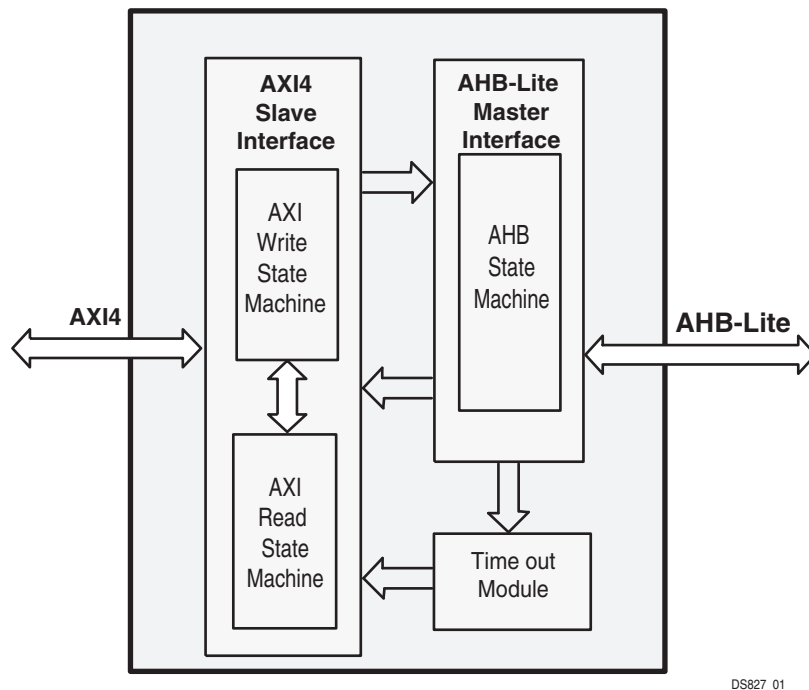


Figure 1: AXI to AHB-Lite Bridge Block Diagram

AXI4 Slave Interface

The AXI4 Slave Interface module provides a bidirectional slave interface. The AXI4 address width is fixed at 32 bits. The AXI4 data bus width can be either 32 or 64 bits, based on the parameter `C_S_AXI_DATA_WIDTH`. The AXI to AHB-Lite Bridge supports the same data width on both the AXI4 and AHB-Lite interfaces.

When both write and read transfers are simultaneously requested on the AXI4 interface, the read request is given a higher priority than the write request.

AXI Write State Machine

The AXI Write State Machine is part of the AXI4 slave interface module and functions on AXI4 write channels. This module controls AXI4 write accesses and generates the write response. If a bridge timeout occurs, this module completes the AXI write transaction with a `SLVERR` response.

AXI Read State Machine

The AXI Read State Machine is part of the AXI4 slave interface module and functions on AXI4 read channels. This module controls the AXI4 read accesses and generates the read response. If a bridge timeout occurs, this module completes the AXI read transaction with a `SLVERR` response.

AHB-Lite Master Interface

The AHB-Lite Master Interface module provides the AHB-Lite master interface. The AHB-Lite address width is fixed at 32 bits and the data bus width can be either 32 or 64 bits, based on the parameter `C_M_AHB_DATA_WIDTH`. `C_M_AHB_DATA_WIDTH` cannot be set by the user, because it is updated automatically by `C_S_AXI_DATA_WIDTH`.

AHB State Machine

The AHB state machine is part of the AHB-Lite master interface module.

When the AXI4 interface initiates a write access, the AHB state machine module receives the control signals and data from AXI4 slave interface, then transfers the same to the equivalent AHB-Lite write access. This module also transfers the AHB-Lite write response to the AXI4 slave interface.

When the AXI4 interface initiates a read access, the AHB state machine module receives the control signals from AXI4 slave interface, then transfers the same to the equivalent AHB-Lite read access. This module also transfers AHB-Lite read data and read response to the AXI4 slave interface.

Timeout Module

The timeout module generates the timeout when the AHB-Lite slave is not responding to the AHB transaction. This is parameterized and generates the timeout only when `C_DPHASE_TIMEOUT` value is non-zero.

The timeout module waits for `C_DPHASE_TIMEOUT` number of AXI clocks for the AHB-Lite slave response and generates the timeout if the AHB slave does not respond.

I/O Signals

Table 1 shows the I/O signals of the AXI to AHB-Lite Bridge.

Table 1: I/O Signal Description

Port	Signal Name	Interface	I/O	Initial State	Description
AXI Interface System Signals					
P1	S_AXI_ACLK	System	I	-	AXI clock.
P2	S_AXI_ARESETN	System	I	-	AXI reset, active-Low.
AXI Write Address Channel Signals					
P3	S_AXI_AWID[C_S_AXI_ID_WIDTH-1:0]	AXI4	I	-	Write address ID: this signal is the identification tag for the write address group of signals.
P4	S_AXI_AWLEN[7:0]	AXI4	I	-	Burst length: the burst length gives the exact number of transfers in a burst.
P5	S_AXI_AWSIZE[2:0]	AXI4	I	-	Burst size: this signal indicates the size of each transfer in the burst.
P6	S_AXI_AWBURST[1:0]	AXI4	I	-	Burst type: the burst type coupled with the size information details how the address for each transfer within the burst is calculated.
P7	S_AXI_AWCACHE[3:0]	AXI4	I	-	Cache type: this signal indicates the bufferable, cacheable, write-through, write-back, and allocate attributes of the transaction.
P8	S_AXI_AWLOCK	AXI4	I	-	Lock type: this signal provides additional information about the atomic characteristics of the transfer.
P9	S_AXI_AWADDR[C_S_AXI_ADDR_WIDTH-1:0]	AXI4	I	-	AXI Write address: the write address bus gives the address of the first transfer in a write burst transaction.
P10	S_AXI_AWPROT[2:0]	AXI4	I	-	Protection type: this signal indicates the normal, privileged, or secure protection level of the write transaction and whether the transaction is a data access or an instruction access. The default value is normal non secure data access.
P11	S_AXI_AWVALID	AXI4	I	-	Write address valid: this signal indicates that valid write address and control information are available.
P12	S_AXI_AWREADY	AXI4	O	0	Write address ready: this signal indicates that the slave is ready to accept an address and associated control signals.
AXI Write Data Channel Signals					
P13	S_AXI_WDATA[C_S_AXI_DATA_WIDTH-1:0]	AXI4	I	-	Write data bus.
P14	S_AXI_WSTRB[C_S_AXI_DATA_WIDTH/8-1:0]	AXI4	I	-	Write strobes: this signal indicates which byte lanes to update in memory.
P15	S_AXI_WVALID	AXI4	I	-	Write valid: this signal indicates that valid write data and strobes are available.
P16	S_AXI_WLAST	AXI4	I	-	Write last: this signal indicates the last transfer in a write burst.
P17	S_AXI_WREADY	AXI4	O	0	Write ready: this signal indicates that the slave can accept the write data.

Table 1: I/O Signal Description (Cont'd)

Port	Signal Name	Interface	I/O	Initial State	Description
AXI Write Response Channel Signals					
P18	S_AXI_BID[C_S_AXI_ID_WIDTH-1:0]	AXI4	O	0	Response ID: the identification tag of the write response.
P19	S_AXI_BRESP[1:0]	AXI4	O	0	Write response: this signal indicates the status of the write transaction.
P20	S_AXI_BVALID	AXI4	O	0	Write response valid: this signal indicates that a valid write response is available.
P21	S_AXI_BREADY	AXI4	I	-	Response ready: this signal indicates that the master can accept the response information.
AXI Read Address Channel Signals					
P22	S_AXI_ARID[C_S_AXI_ID_WIDTH -1:0]	AXI4	I	-	Read address ID: this signal is the identification tag for the read address group of signals.
P23	S_AXI_ARADDR[C_S_AXI_ADDR_WIDTH -1:0]	AXI4	I	-	Read address: the read address bus gives the initial address of a read burst transaction.
P24	S_AXI_ARCACHE[3:0]	AXI4	I	-	Cache type: this signal provides additional information about the cacheable characteristics of the transfer.
P25	S_AXI_ARPROT[2:0]	AXI4	I	-	Protection type: this signal provides protection unit information for the read transaction. The default value is normal non secure data access.
P26	S_AXI_ARVALID	AXI4	I	-	Read address valid: this signal indicates when High that the read address and control information is valid and remains stable until the address acknowledgement signal S_AXI_ARREADY is High.
P27	S_AXI_ARLEN[7:0]	AXI4	I	-	Burst length: the burst length gives the exact number of transfers in a burst.
P28	S_AXI_ARSIZE[2:0]	AXI4	I	-	Burst size: this signal indicates the size of each transfer in the burst.
P29	S_AXI_ARBURST[1:0]	AXI4	I	-	Burst type: the burst type coupled with the size information details how the address for each transfer within the burst is calculated.
P30	S_AXI_ARLOCK	AXI4	I	-	Lock type: this signal provides additional information about the atomic characteristics of the transfer.
P31	S_AXI_ARREADY	AXI4	O	0	Read address ready: this signal indicates that the slave is ready to accept an address and associated control signals.
AXI Read Data Channel Signals					
P32	S_AXI_RID[C_S_AXI_ID_WIDTH -1:0]	AXI4	O	0	Read ID tag: this signal is the identification tag for the read data group of signals.
P33	S_AXI_RDATA[C_S_AXI_DATA_WIDTH -1:0]	AXI4	O	0	Read data bus.
P34	S_AXI_RLAST	AXI4	O	0	Read last: this signal indicates the last transfer in a read burst.
P35	S_AXI_RRESP[1:0]	AXI4	O	0	Read response: this signal indicates the status of the read transfer.

Table 1: I/O Signal Description (Cont'd)

Port	Signal Name	Interface	I/O	Initial State	Description
P36	S_AXI_RVALID	AXI4	O	0	Read valid: this signal indicates that the required read data is available and the read transfer can complete.
P37	S_AXI_RREADY	AXI4	I	-	Read ready: this signal indicates that the master can accept the read data and response information.
AHB-Lite Signals					
P38	M_AHB_HCLK	AHB-Lite	O	-	AHB Clock - S_AXI_ACLK is tied to M_AHB_HCLK
P39	M_AHB_HRESETN	AHB-Lite	O	-	AHB Reset, active-Low - S_AXI_ARESETN is tied to M_AHB_HRESETN.
P40	M_AHB_HADDR[C_M_AHB_ADDR_WIDTH -1:0]	AHB-Lite	O	0	This is the AHB address bus and is fixed to 32-bit.
P41	M_AHB_HPROT[3:0]	AHB-Lite	O	0011 ⁽¹⁾	Protection type: this signal indicates the normal, privileged transaction and whether the transaction is a data access or an instruction access.
P42	M_AHB_HTRANS[1:0]	AHB-Lite	O	0	AHB Transfer Type, which can be NONSEQUENTIAL, SEQUENTIAL, IDLE or BUSY.
P43	M_AHB_HSIZE[2:0]	AHB-Lite	O	0	AHB Size of Transfer.
P44	M_AHB_HWRITE	AHB-Lite	O	0	Indicates the transfer direction, this signal indicates an AHB write access when High and an AHB read access when Low.
P45	M_AHB_HWDATA[C_M_AHB_DATA_WIDTH -1:0]	AHB-Lite	O	0	Write data: The write data bus transfers data from the master to the slaves during write operations.
P46	M_AHB_HBURST[2:0]	AHB-Lite	O	0	AHB Burst type: the burst type indicates if the transfer is a single transfer or forms part of a burst. The burst can be incrementing or wrapping.
P47	M_AHB_HMASTLOCK	AHB-Lite	O	0	Indicates that the current master is performing a locked sequence of transfers
P48	M_AHB_HREADY	AHB-Lite	I	-	Ready: The AHB slave uses this signal to extend an AHB transfer.
P49	M_AHB_HRDATA[C_M_AHB_DATA_WIDTH -1:0]	AHB-Lite	I	-	AHB read data driven by slave.
P50	M_AHB_HRESP	AHB-Lite	I	-	Transfer Response: When Low, indicates that the transfer status is OKAY. When High, indicates that the transfer status is ERROR.

Notes:

1. The default value 0011 is driven on M_AHB_HPROT[3:0] as per the ARM® recommendation, this corresponds to non-cacheable, non-bufferable, privileged, data access.

Design Parameters

Table 2 shows the design parameters of the AXI to AHB-Lite Bridge. In addition to the parameters listed in Table 2, there are also parameters that are inferred for each AXI interface in the EDK tools. Through the design, these EDK-inferred parameters control the behavior of the AXI Interconnect. For a complete list of the interconnect settings related to the AXI interface, see the *LogiCORE AXI Interconnect IP Data Sheet* [Ref 6].

Table 2: Design Parameters

Generic	Feature/Description	Parameter Name	Allowable Values	Default Values	VHDL Type
System Parameter					
G1	Target FPGA family	C_FAMILY	zynq, virtex7, kintex7, artix7, virtex6, spartan6	virtex6	string
AXI Interconnect Parameters					
G2	AXI interface type	C_S_AXI_PROTOCOL	axi4	axi4 ⁽¹⁾	string
G3	AXI write support	C_S_AXI_SUPPORTS_WRITE	1	1 ⁽¹⁾	integer
G4	AXI read support	C_S_AXI_SUPPORTS_READ	1	1 ⁽¹⁾	integer
G5	AXI write acceptance	C_S_AXI_WRITE_ACCEPTANCE	1	1 ⁽¹⁾	integer
G6	AXI read acceptance	C_S_AXI_READ_ACCEPTANCE	1	1 ⁽¹⁾	integer
AXI Parameters					
G7	AXI address bus width	C_S_AXI_ADDR_WIDTH	32	32	integer
G8	AXI data bus width	C_S_AXI_DATA_WIDTH	32,64 ⁽²⁾	32	integer
G9	AXI Narrow burst support	C_S_AXI_SUPPORTS_NARROW_BURST	0,1	0	integer
G10	AXI ID width	C_S_AXI_ID_WIDTH	1-16	4	integer
AHB-Lite Parameters					
G11	AHB-Lite address bus width	C_M_AHB_ADDR_WIDTH	32	32	integer
G12	AHB-Lite data bus width	C_M_AHB_DATA_WIDTH	32,64 ⁽²⁾	32	integer
AXI to AHB-Lite Bridge Parameters					
G13	Timeout value in AXI clocks	C_DPHASE_TIMEOUT	0,16,32,64,128,256 ⁽³⁾	0	integer

Notes:

- These generics are needed by the system. This bridge supports both read and write. The AXI4 interface only is supported by AXI to AHB-Lite Bridge.
- The AXI4 data width and AHB-Lite data width should be same. C_M_AHB_DATA_WIDTH cannot be set by the user and is modified automatically; the same as is done with C_S_AXI_DATA_WIDTH. If C_S_AXI_DATA_WIDTH is 32, then C_M_AHB_DATA_WIDTH is also 32; if C_S_AXI_DATA_WIDTH is 64, then C_M_AHB_DATA_WIDTH is also 64.
- The timeout module is parameterized and does not generate the timeout if the C_DPHASE_TIMEOUT value is set to 0.

Parameter - I/O Signal Dependencies

The dependencies between the AXI to AHB-Lite Bridge core design parameters and the I/O signals are described in Table 3. In addition, when certain features are parameterized out of the design, the related logic is no longer a part of the design. In the AXI to AHB-Lite Bridge core, narrow transfer support and timeout logic are parameterized. When the C_S_AXI_SUPPORTS_NARROW_BURST is 1, only the related narrow burst support logic is part of the

design. Similarly, when C_DPHASE_TIMEOUT is a non-zero integer (for example, 16/32/64/128/256), the corresponding timeout logic exists in the design.

Table 3: Parameter-I/O Signal Dependencies

Generic or Port	Name	Affects	Depends	Relationship Description
Design Parameters				
G8	C_S_AXI_DATA_WIDTH	P13,P14, P33, G14		Affects the AXI data bus width and also AHB-Lite data bus width
G10	C_S_AXI_ID_WIDTH	P3,P18, P22,P32	-	Affects the width of the signals S_AXI_AWID, S_AXI_BID, S_AXI_ARID and S_AXI_RID
G12	C_M_AHB_DATA_WIDTH	P45,P49	G8	Affects the AHB-Lite data bus width
I/O Signals				
P3	S_AXI_AWID[C_S_AXI_ID_WIDTH-1:0]	-	G10	This signal width is affected based on the C_S_AXI_ID_WIDTH value
P13	S_AXI_WDATA[C_S_AXI_DATA_WIDTH-1:0]	-	G8	This signal width is 32/64 based on C_S_AXI_DATA_WIDTH
P14	S_AXI_WSTRB[C_S_AXI_DATA_WIDTH/8-1:0]	-	G8	This write data strobe signal width is 4/8 based on C_S_AXI_DATA_WIDTH
P18	S_AXI_BID[C_S_AXI_ID_WIDTH-1:0]	-	G10	This signal width is affected based on the C_S_AXI_ID_WIDTH value
P22	S_AXI_ARID[C_S_AXI_ID_WIDTH-1:0]	-	G10	This signal width is affected based on the C_S_AXI_ID_WIDTH value
P32	S_AXI_RID[C_S_AXI_ID_WIDTH-1:0]	-	G10	This signal width is affected based on the C_S_AXI_ID_WIDTH value
P33	S_AXI_RDATA[C_S_AXI_DATA_WIDTH-1:0]	-	G8	This signal width is 32/64 based on C_S_AXI_DATA_WIDTH
P45	M_AHB_HWDATA[C_M_AHB_DATA_WIDTH-1:0]	-	G12	This signal width is 32/64 based on C_M_AHB_DATA_WIDTH
P49	M_AHB_HRDATA[C_M_AHB_DATA_WIDTH-1:0]	-	G12	This signal width is 32/64 based on C_M_AHB_DATA_WIDTH

Design Details

Clocking

The AXI to AHB-Lite Bridge is a synchronous design which uses the S_AXI_ACLK at both the AXI and AHB-Lite interfaces. M_AHB_HCLK is driven by the AXI to AHB-Lite Bridge (tied to S_AXI_ACLK).

Reset

S_AXI_ARESETN is a synchronous reset input that resets the AXI to AHB-Lite Bridge when asserted. The S_AXI_ARESETN is also used to reset the AHB-Lite interface. M_AHB_HRESETN is driven by the AXI to AHB-Lite Bridge (tied to S_AXI_ARESETN).

Memory Mapping

The AXI memory map and the AHB-Lite memory map are a single, complete 32-bit (4 GB) memory space. The AXI to AHB-Lite Bridge does not modify the address for AHB-Lite; therefore, the address that is presented on the AHB-Lite is exactly as received on the AXI interface.

Data Width

The AXI and AHB-Lite data widths are either 32 or 64, based on the parameters, C_S_AXI_DATA_WIDTH and C_M_AHB_DATA_WIDTH. The AXI to AHB-Lite Bridge supports the same data widths on both AXI and AHB-Lite interfaces. The parameters, C_S_AXI_DATA_WIDTH and C_M_AHB_DATA_WIDTH should be the same. When the user sets the C_S_AXI_DATA_WIDTH to 32/64, the same value is automatically applied to C_M_AHB_DATA_WIDTH.

Narrow Transfers

Transactions where the transfer size is narrower than the data bus width are treated as narrow transfers. Narrow transfer support is parameterized in the AXI to AHB-Lite Bridge. Narrow transfers on the AXI4 interface are supported and can be transferred to and from AHB-Lite interface when C_S_AXI_SUPPORTS_NARROW_BURST is set to 1.

8 and 16 bit data transfers are allowed in addition to the 32-bit data transfers, when C_S_AXI_SUPPORTS_NARROW_BURST is 1 and C_M_AHB_DATA_WIDTH and C_S_AXI_DATA_WIDTH are 32. When C_S_AXI_SUPPORTS_NARROW_BURST is 0 and C_M_AHB_DATA_WIDTH and C_S_AXI_DATA_WIDTH are 32, then 32-bit data transfers only are allowed.

The 8/16/32-bit data transfers are allowed in addition to the 64-bit transfers when C_S_AXI_SUPPORTS_NARROW_BURST is 1 and C_M_AHB_DATA_WIDTH and C_S_AXI_DATA_WIDTH are 64. When C_S_AXI_SUPPORTS_NARROW_BURST is 0 and C_M_AHB_DATA_WIDTH and C_S_AXI_DATA_WIDTH are 64, then 64-bit data transfers only are allowed. The AXI to AHB-Lite Bridge monitors the AXI strobe, S_AXI_WSTRB, and drives M_AHB_HSIZE when there are single transfers initiated by the AXI master.

The AXI strobe, S_AXI_WSTRB, is ignored in the AXI to AHB-Lite Bridge for burst transfers. Also, for burst transfers, M_AHB_HSIZE is determined based on S_AXI_AWADDR/S_AXI_ARADDR and S_AXI_AWSIZE/S_AXI_ARSIZE. Sparse/unaligned transfers are not supported in burst transfers.

Endianness

Both the AXI and AHB-Lite are little-endian.

Address/data translation

No address/data translation/conversion from AXI4 to AHB-Lite takes place inside the AXI to AHB-Lite Bridge. The write/read address from AXI4 is passed to AHB-Lite address. AXI4 write data is passed on to AHB-Lite and AHB-Lite read data is passed on to AXI4 read data.

Read and Write Ordering

When a read and a write request are issued simultaneously (S_AXI_AWVALID/S_AXI_WVALID and S_AXI_ARVALID are asserted High) from the AXI4 interface, the AXI to AHB-Lite Bridge gives a higher priority to

the read request. When both write and read requests are valid, the write request is initiated on AHB-Lite after the read is requested on AHB-Lite interface.

1 kB Burst Crossing

The AXI to AHB-Lite Bridge implements the logic that checks if the AXI transaction (both read and write) crosses the 1 kB boundary, then splits the transaction to prevent it from crossing boundaries. Any burst initiated on the AXI interface that crosses a 1 kB address boundary is converted to two INCR bursts of undefined length with the restarted burst on AHB-Lite.

Bridge Timeout Condition

Timeout logic is parameterized in the AXI to AHB-Lite Bridge. A timeout is generated when the parameter C_DPHASE_TIMEOUT value is 16, 32, 64, 128 or 256. When a request is issued from the AXI interface, the bridge translates this request into the corresponding AHB-Lite transfer and requests it on the AHB-Lite interface. If this request is not responded to by the AHB-Lite interface, the timeout logic waits for a timeout period and responds with SLVERR on the AXI interface. The bridge sends an IDLE transfer to AHB-Lite interface.

If the C_DPHASE_TIMEOUT parameter value is 0, it is assumed that the AHB slave always responds when a transfer is requested and that no timeout logic exists that automatically responds on the AXI interface. When the AHB slave does not respond, the AXI to AHB-Lite Bridge waits indefinitely for the AHB-Lite slave response.

Note: Select the C_DPHASE_TIMEOUT value to avoid the AXI to AHB-Lite Bridge waiting indefinitely for the AHB-Lite slave response.

AXI Response Signaling

Only OKAY and SLVERR responses are generated on the AXI side. EXOKAY and DECERR are never used. An ERROR on the AHB-Lite interface or a timeout error because of a bridge timeout results in SLVERR.

No registers are implemented in AXI to AHB-Lite Bridge to differentiate the AHB-Lite ERROR and bridge timeout error. It is assumed and recommended that the AXI master resets the bridge with SLVERR response.

Protection and Cache Support

The protection and cache support is limited in the AXI to AHB-Lite Bridge. The protection and cache support in AXI4 is mapped to the available equivalent AHB-Lite protection as listed in the [Table 4](#). In AXI4, no cacheable transaction exists in the AHB-Lite. There are no modifiable, write allocate, read allocate, secure and non secure accesses in the AHB-Lite interface that exist in AXI4 interface.

The protection support privileged, instruction, normal, and data accesses in AXI4 are mapped to the equivalent AHB-Lite protection. The S_AXI_AWPROT[2:0] and S_AXI_ARPROT[2:0] signals carry the protection unit support on the AXI4 interface.

The AXI4 bufferable/non-bufferable transaction attributes are mapped to the equivalent AHB-Lite protection signals. The S_AXI_AWCACHE [3:0] and S_AXI_ARCACHE[3:0] carry the cache support attributes of AXI4.

Table 4: AXI4 Protection And Cache Support Translation to AHB-Lite Protection

AXI4 Protection and Cache Support		AHB-Lite Protection Support	Description
S_AXI_AWCACHE[3:0]/ S_AXI_ARCACHE[3:0]	S_AXI_AWPROT[2:0]/ S_AXI_ARPROT[2:0]	M_AHB_HPROT[3:0]	
xxxx	xx1	0x1x	Privileged on AXI4 is translated to the equivalent privileged in AHB-Lite
xxxx	xx0	0x0x	Normal access on AXI is mapped to the equivalent user access on AHB
xxxx	0xx	0xx1	Data access on AXI4 is translated to the equivalent Data access in AHB-Lite
xxxx	1xx	0xx0	Instruction access on AXI4 is mapped to the equivalent Opcode fetch on AHB
00x1	xxx	01xx	Bufferable on AXI4 is translated to the equivalent Bufferable in AHB-Lite
00x0	xxx	00xx	Non-bufferable on AXI4 is translated to the equivalent Non-bufferable in AHB-Lite
xxxx	xxx	0xxx	There is no cacheable in AXI4. Always transferring Non-cacheable on AHB-Lite

As per the ARM recommendation [Ref 2] the default value on AHB-Lite protection control signal M_AHB_HPROT[3:0] is 0011 to correspond to a non-cacheable, non-bufferable, privileged data access.

Register Descriptions

There are no registers in the AXI to AHB-Lite Bridge.

Bridge Transaction Translation

Table 5 shows translation of AXI4 transactions to AHB-Lite transactions. The supported AXI4 transactions are translated to AHB-Lite transactions.

- Incrementing burst of length 1 on AXI4 is converted to the equivalent AHB single transaction
- Incrementing burst transfers of length 4, 8, and 16 on AXI4 are converted to the equivalent AHB-Lite incrementing bursts when there is no 1 kB cross in the access
- Any other incrementing burst of length up to 256 (2 to 256, other than 4, 8, and 16) is converted to the AHB-Lite incrementing burst transfer of undefined length
- Wrapping burst transfers of length 4, 8, and 16 are converted to the equivalent AHB-Lite wrapping burst

For AXI4 transactions given below, multiple AHB-Lite transactions must be performed.

- For one AXI4 transaction, two AHB-Lite transactions must be requested when a 1 kB cross is detected in an AXI4 transfer. If there is 1 kB cross access with incrementing burst transfers of length 4, 8, and 16, then the corresponding transaction on AXI4 is converted to the two AHB-Lite incrementing burst transfers of undefined length
- AXI4 allows WRAP type burst transactions of length 2, 4, 8, and 16; however, the AHB-Lite interface only supports WRAP 4, 8, and 16 transactions. WRAP transfer of length 2 on the AXI4 interface is converted to two AHB-Lite single burst transactions

- Fixed burst is supported in AXI4 whereas there is no fixed burst transaction in AHB-Lite. Fixed burst on AXI4 is converted to AHB-Lite single transactions with fixed address.

Table 5: AXI4 Transaction to AHB-Lite Transaction

AXI4 Transaction	AHB-Lite Transaction	Description
Incrementing burst transaction of length 1	Single transaction	AXI4 incrementing burst transaction with length 1 is single transaction on AHB
Incrementing burst transfers of length 4, 8, and 16 without one Kb crossing	Incrementing bursts 4, 8, and 16	If the access initiated by AXI does not cross the 1 kB, INCR4, INCR8, and INCR16 are same in both AXI and AHB-Lite
Incrementing burst transfers of length 4, 8, and 16 with one kb crossing	Incrementing burst transfers of undefined length	If the access initiated by AXI crosses the 1 kB, the transfer is split so the transfers on AHB-Lite are two increment bursts of undefined length transfers.
Any other incrementing burst up to 256 (Incrementing bursts of length from 2 to 256 other than 4, 8, and 16)	Incrementing burst transfers of undefined length	If the access is incrementing burst of length 2 to 256 other than 4, 8, and 16, the transfer on AHB-Lite is incrementing burst of undefined length transfer.
WRAP type burst transactions of length 4, 8, and 16	WRAP type burst transactions of 4, 8, and 16	WRAP4, WRAP8, and WRAP16 are the same in both AXI and AHB
WRAP type burst transaction of length 2	Two single transactions	There is no WRAP2 burst transfer in AHB-Lite. The transaction is split as two, single AHB-Lite transactions
Fixed burst transactions of length 1 to 16	Single transactions with the same address	There is no fixed burst transaction in AHB-Lite. Fixed transaction on AXI4 is single transactions on AHB-Lite to the same address location. The number of single transactions (for example, 1 to 16) depends on the AXI burst length.

Un-supported Features/Limitations

The following are not supported:

AXI4 Slave Interface

- Data bus widths greater than 64
- No registers are implemented because posted writes
- Locked, Barrier, trust zone, and exclusive operations
- Out-of-order read transaction completion
- Out-of-order write transaction completion
- Unaligned/Sparse burst transfers (holes in strobes)
- EXOKAY and DECERR responses to AXI4
- Low-power state
- Secure accesses

AHB-Lite Master interface

- Data bus widths greater than 64
- Cacheable access

Timing Diagrams

The AXI to AHB-Lite Bridge operation for various read and write transfers is shown in the subsequent timing diagrams.

1. AXI incrementing burst read transfer of length 1 is shown in Figure 2.
2. AXI incrementing burst write transfer of length 1 is shown in Figure 3.
3. AXI read and write transfers is shown in Figure 4. As shown, when read and write transfers are initiated on the AXI4 interface at the same time, the read transfer is given a higher priority than the write transfer.
4. AXI incrementing burst write transfer of length 4 is shown in Figure 5.
5. AXI incrementing burst read transfer of length 4 is shown in Figure 6.
6. AXI fixed burst read and write transfers are shown in Figure 7. AXI fixed burst read and write transfers of length 5 are transferred to five AHB-Lite single read and write transfers.
7. AXI wrapping burst read and write transfers of length 2 are shown in Figure 8. One AXI wrapping burst read/write transfer of length 2 is split into two AHB-Lite single read/write transfers.
8. AXI wrapping burst read transfer of length 4 is shown in Figure 9.
9. AXI wrapping burst write transfer of length 4 is shown in Figure 10.
10. AXI 8-bit narrow read transfer on 32-bit data bus is shown in Figure 11.
11. AXI 8-bit narrow write transfer on 64-bit data bus is shown in Figure 12.
12. AXI burst read transfer of length 4 that crosses the 1 kB address on the AHB-Lite is shown in Figure 13. One AXI read transfer is split into two AHB-Lite read transfers as the 1 kB boundary is crossed.
13. AXI burst write transfer of length 4 that crosses the 1 kB Address on the AHB-Lite is shown in Figure 14. One AXI write transfer is split into two AHB-Lite write transfers as the 1 kB boundary is crossed.
14. AXI read timeout when C_DPHASE_TIMEOUT is 16 is shown in Figure 15. AXI read transfer is completed with a SLVERR response.
15. AXI write timeout when C_DPHASE_TIMEOUT is 16 is shown in Figure 16. AXI write transfer is completed with a SLVERR response.

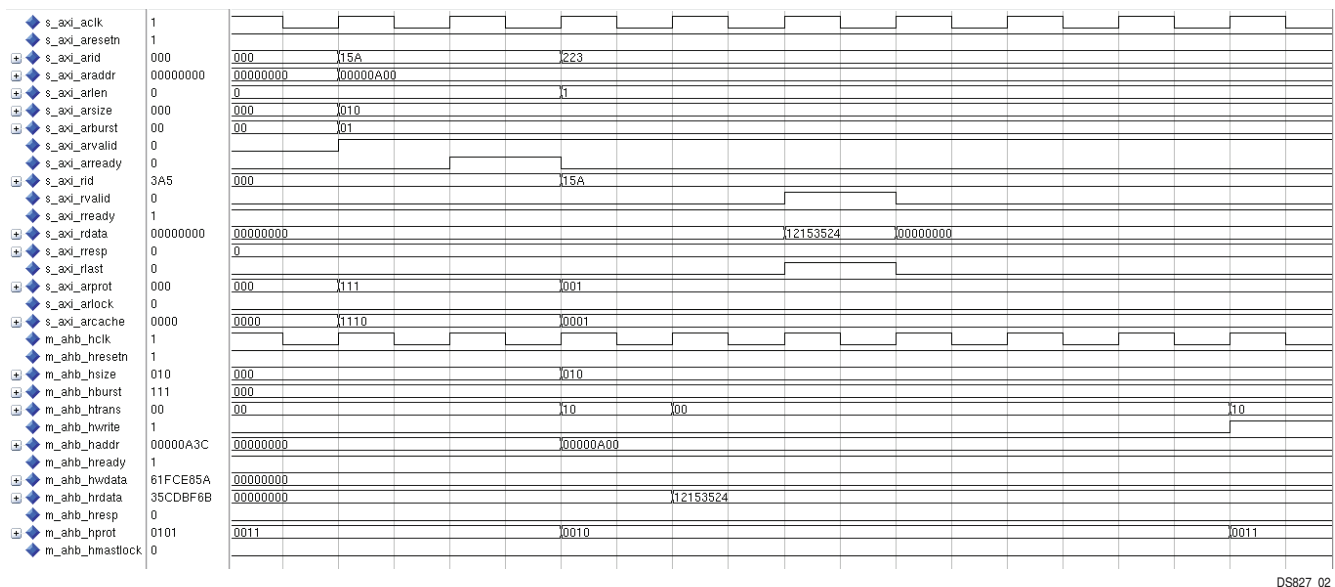


Figure 2: AXI Incrementing Burst Read Transfer of Length 1

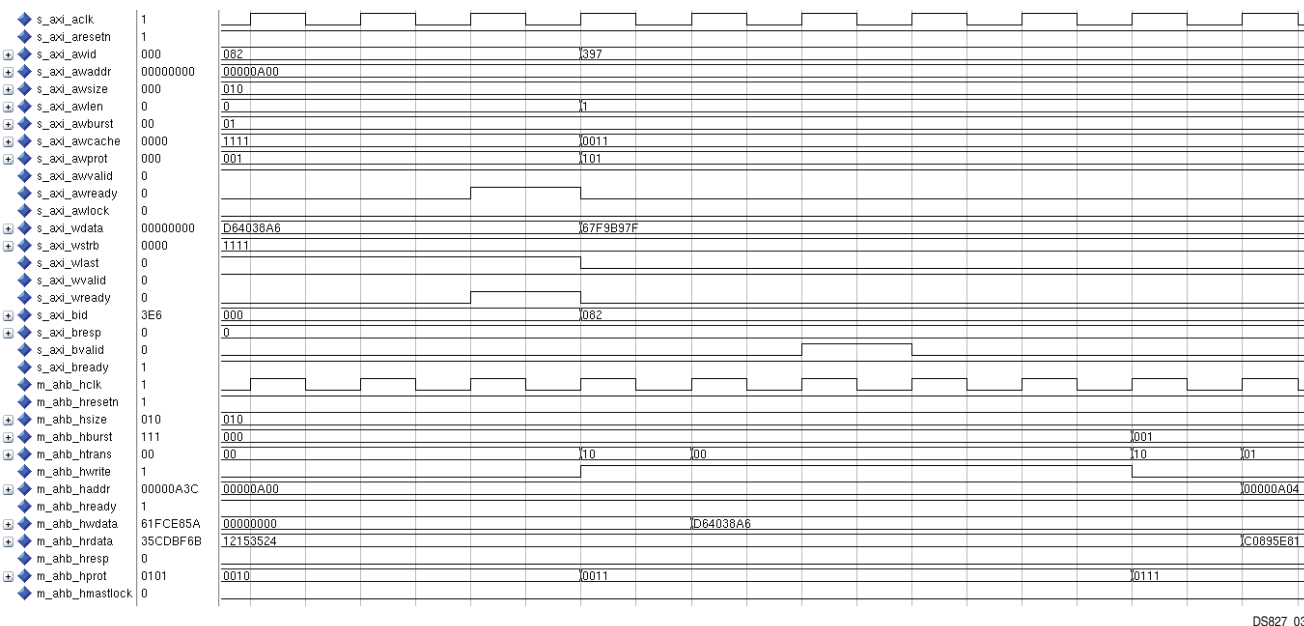


Figure 3: AXI Incrementing Burst Write Transfer of Length 1

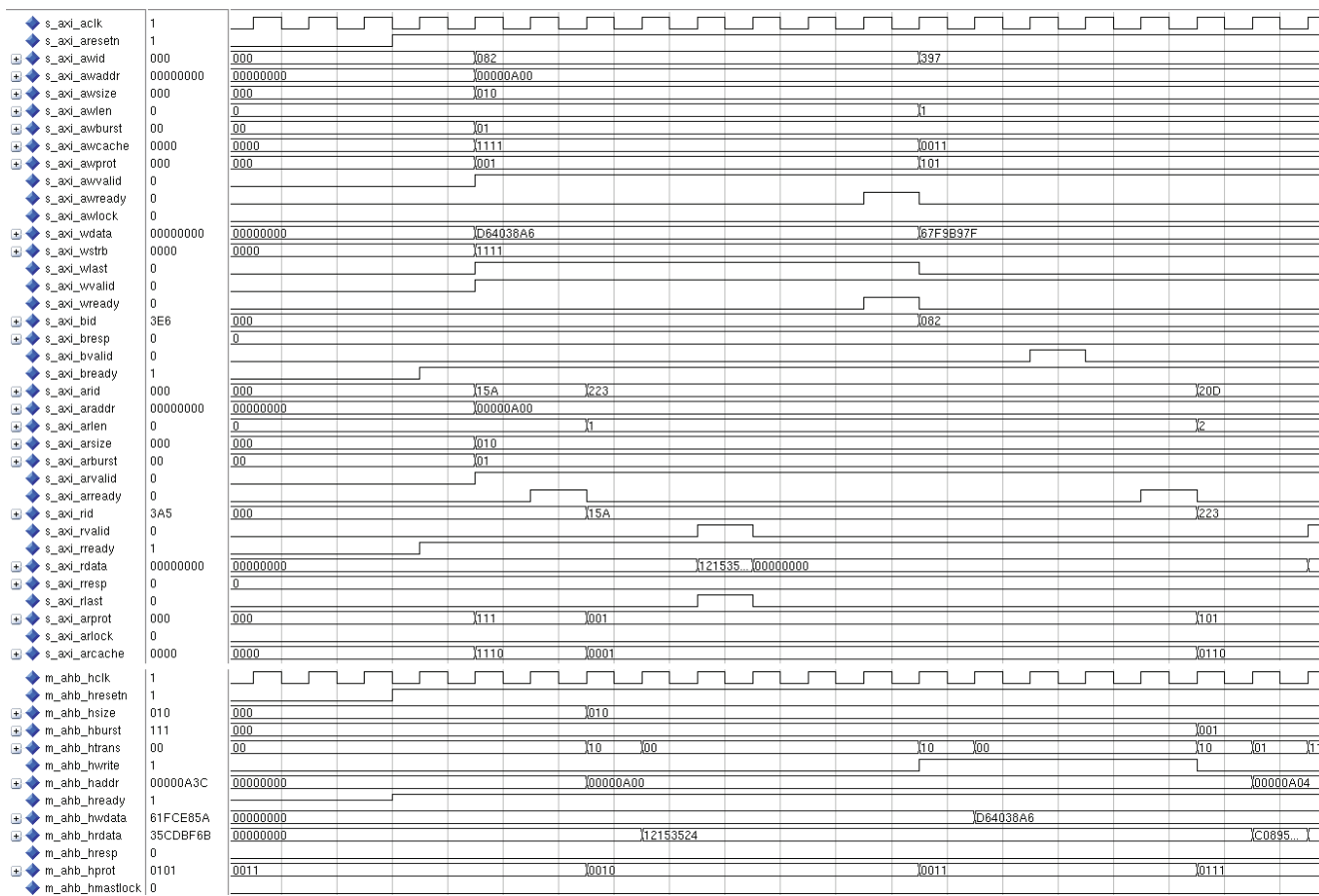


Figure 4: AXI Read and Write Transfers



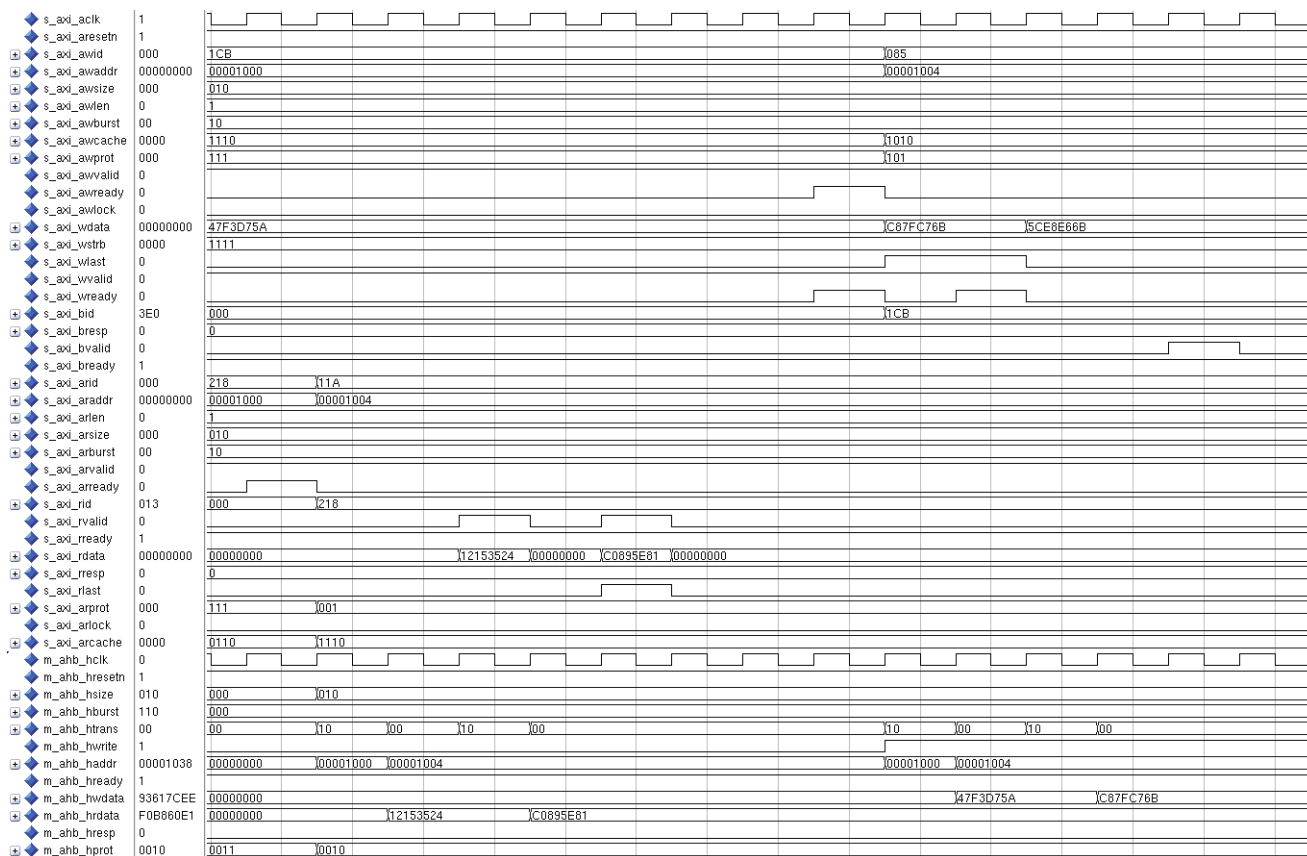
Figure 5: **AXI Incrementing Burst Write Transfer of Length 4**



Figure 6: AXI Incrementing Burst Read Transfer of Length 4

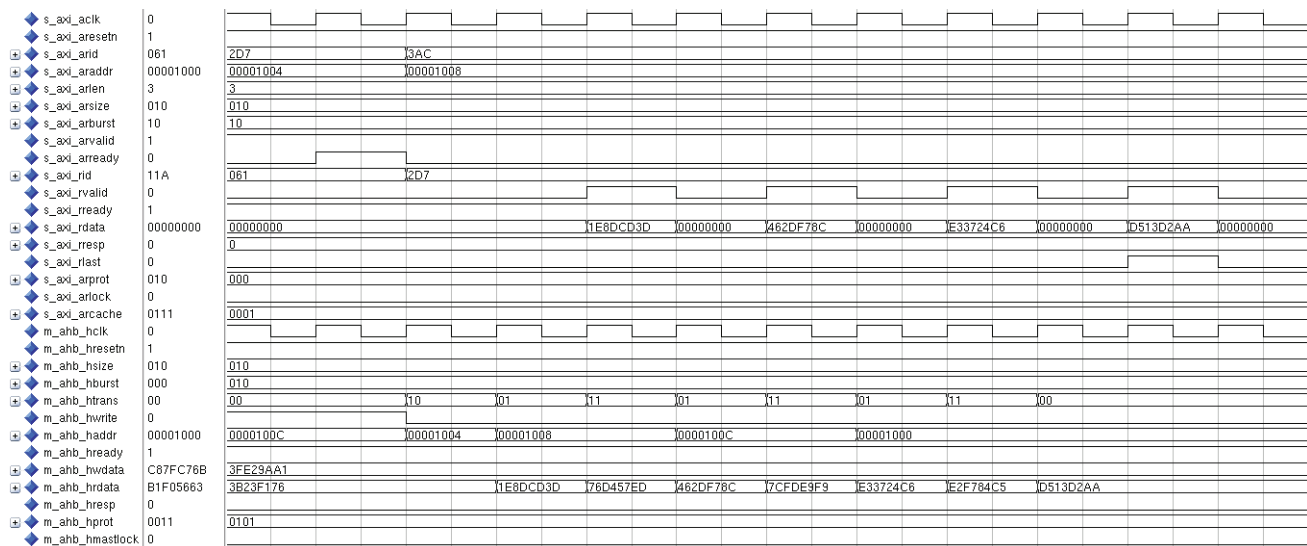


Figure 7: AXI Fixed Burst Read and Write Transfers



DS827_08

Figure 8: AXI Wrapping Burst Read and Write Transfer of Length 2



DS827_09

Figure 9: AXI Wrapping Burst Read Transfer of Length 4



Figure 10: AXI Wrapping Burst Write Transfer of Length 4



Figure 11: AXI 8-bit Narrow Read Transfer on 32-bit Data Bus



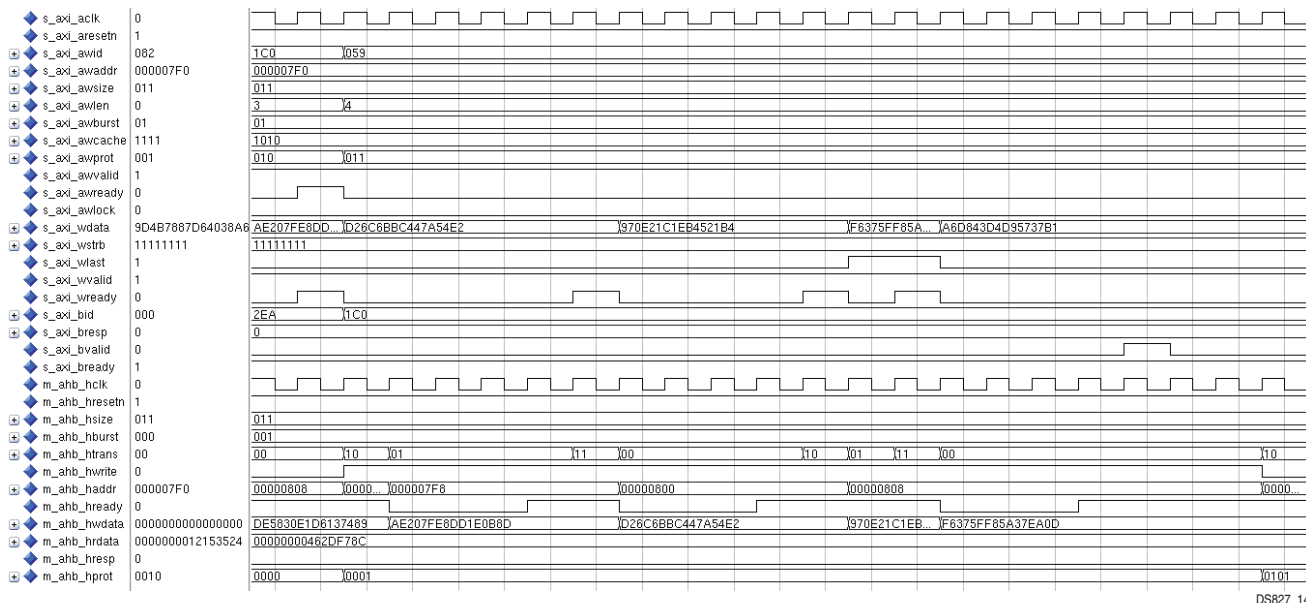


Figure 14: AXI Burst Write Transfer of Length 4 That Crosses 1 kB Address Boundary on AHB-Lite

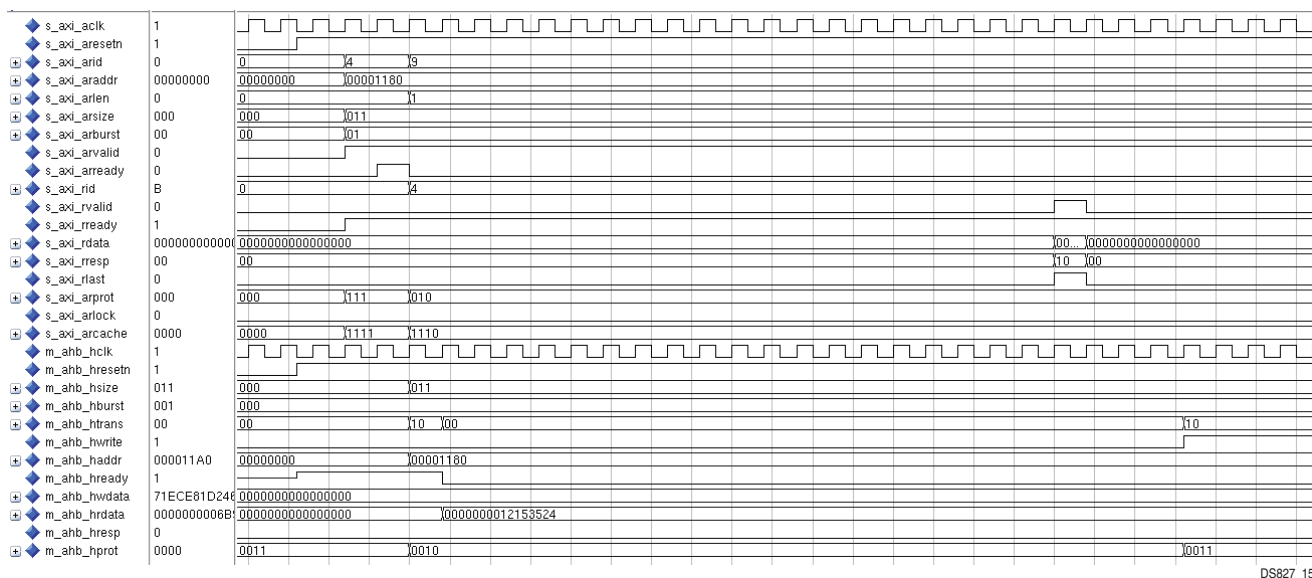


Figure 15: AXI Read Timeout When C_DPHASE_TIMEOUT is 16

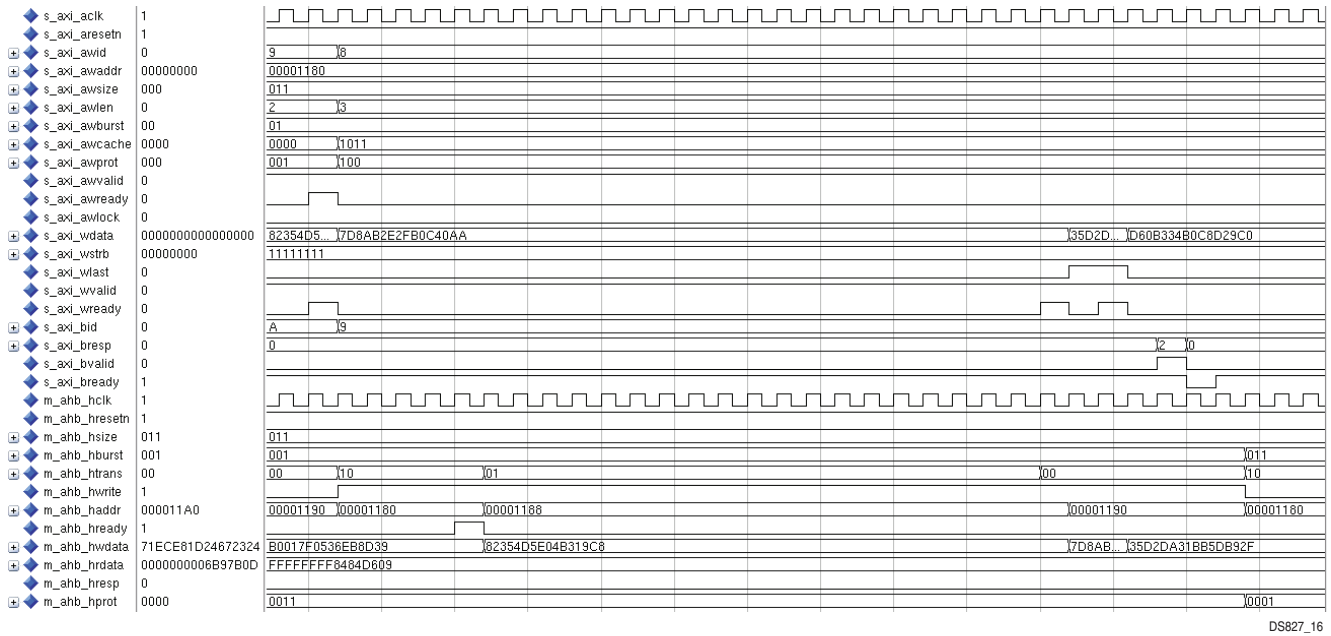


Figure 16: AXI Write Timeout When C_DPHASE_TIMEOUT is 16

DS827_16

Device Utilization and Performance Benchmarks

Core Performance

Because the AXI to AHB-Lite Bridge module is used with other designs in the FPGA, the resource utilization and timing numbers reported in this section are estimates only.

The AXI to AHB-Lite Bridge resource utilization benchmarks for several parameter combinations measured with Virtex-6 FPGA as the target device are shown in [Table 6](#).

Table 6: Performance and Resource Utilization Benchmarks for Virtex-6 FPGA⁽¹⁾

Parameter Values (Other parameters at default value)					Device Resources			Performance
C_S_AXI_DATA_WIDTH	C_M_AHB_DATA_WIDTH	C_S_AXI_ID_WIDTH	C_DPHASE_TIMEOUT	C_S_AXI_SUPPORTS_NARROW_BURST	Slices	Slice Flip-Flops	LUTs	FMAX (MHz)
32	32	4	0	0	176	430	291	215
32	32	4	0	1	198	434	306	210
64	64	4	0	0	222	585	336	213
64	64	16	0	1	238	663	351	208
64	64	16	0	0	230	657	360	223
64	64	16	128	0	235	670	321	201
64	64	4	256	1	223	601	339	218

1. Virtex-6 PPGA (xc6vlx130t-ff1156-1)

The AXI to AHB-Lite Bridge resource utilization benchmarks for several parameter combinations measured with Spartan-6 FPGA as the target device are shown in [Table 7](#).

Table 7: Performance and Resource Utilization Benchmarks for Spartan-6 FPGA⁽¹⁾

Parameter Values(Other parameters at default value)					Device Resources			Performance
C_S_AXI_DATA_WIDTH	C_M_AHB_DATA_WIDTH	C_S_AXI_ID_WIDTH	C_DPHASE_TIMEOUT	C_S_AXI_SUPPORTS_NARROW_BURST	Slices	Slice Flip-Flops	LUTs	FMAX (MHz)
32	32	4	0	0	190	434	318	184
32	32	4	0	1	184	432	332	175
64	64	4	0	0	252	596	406	173
64	64	16	0	1	295	676	383	171
64	64	16	0	0	250	668	346	177
64	64	16	128	0	246	672	332	175
64	64	4	256	1	264	612	426	175

1. Spartan-6 FPGA (xc6slx100t-fgg900-2)

The AXI to AHB-Lite Bridge resource utilization benchmarks for several parameter combinations measured with Virtex-7 FPGA as the target device are shown in [Table 8](#).

Table 8: Performance and Resource Utilization Benchmarks for Virtex-7 FPGA⁽¹⁾

Parameter Values(Other parameters at default value)					Device Resources			Performance
C_S_AXI_DATA_WIDTH	C_M_AHB_DATA_WIDTH	C_S_AXI_ID_WIDTH	C_DPHASE_TIMEOUT	C_S_AXI_SUPPORTS_NARROW_BURST	Slices	Slice Flip-Flops	LUTs	FMAX (MHz)
32	32	4	0	0	182	427	287	202
32	32	4	0	1	179	428	295	206
64	64	4	0	0	216	584	338	200
64	64	16	0	1	239	662	356	200
64	64	16	0	0	247	656	346	201
64	64	16	128	0	226	667	336	201
64	64	4	256	1	219	598	344	202

1. Virtex-7 PPGA (xc7v855tffg1157-3))

The AXI to AHB-Lite Bridge resource utilization benchmarks for several parameter combinations measured with Kintex-7 FPGA as the target device are shown in [Table 9](#).

Table 9: Performance and Resource Utilization Benchmarks for Kintex-7 FPGA⁽¹⁾ and Zynq-7000 Devices

Parameter Values(Other parameters at default value)					Device Resources			Performance
C_S_AXI_DATA_WIDTH	C_M_AHB_DATA_WIDTH	C_S_AXI_ID_WIDTH	C_DPHASE_TIMEOUT	C_S_AXI_SUPPORTS_NARROW_BURST	Slices	Slice Flip-Flops	LUTs	FMAX (MHz)
32	32	4	0	0	155	427	299	200
32	32	4	0	1	148	428	327	200
64	64	4	0	0	216	584	326	200
64	64	16	0	1	242	662	340	200
64	64	16	0	0	250	656	326	200
64	64	16	128	0	226	667	305	200
64	64	4	256	1	225	598	327	200

1. Kintex FPGA (xc7k410tffg676-3)

The AXI to AHB-Lite Bridge resource utilization benchmarks for several parameter combinations measured with Artix-7 FPGA as the target device are shown in [Table 10](#).

Table 10: Performance and Resource Utilization Benchmarks for Artix-7 FPGA⁽¹⁾ and Zynq-7000 Devices

Parameter Values(Other parameters at default value)					Device Resources			Performance
C_S_AXI_DATA_WIDTH	C_M_AHB_DATA_WIDTH	C_S_AXI_ID_WIDTH	C_DPHASE_TIMEOUT	C_S_AXI_SUPPORTS_NARROW_BURST	Slices	Slice Flip-Flops	LUTs	FMAX (MHz)
32	32	4	0	0	124	258	305	200
32	32	4	0	1	116	260	302	200
64	64	4	0	0	146	353	308	200
64	64	16	0	1	230	404	318	200
64	64	16	0	0	240	656	356	200
64	64	16	128	0	215	667	356	200
64	64	4	256	1	157	368	390	200

1. Artix-7 FPGA (xc7a350tfbg676-3)

Read and Write Latency

The best possible core configuration has been selected for calculating the read and write latency for the increment, wrapping, and fixed type of burst transfers. The read latency from read address valid (S_AXI_ARVALID) to the data beat (S_AXI_RVALID) of AXI to AHB-Lite Bridge is 4 clock cycles. The write latency from write address valid (S_AXI_AWVALID) to the availability of valid data on AHB data bus is 3 clock cycles.

Reference Documents

The listed documents contain reference information important to understanding the AXI to AHB-Lite Bridge design:

1. AXI4 AMBA AXI Protocol Version: 2.0 Specification
2. AMBA AHB-Lite protocol Version: 1.0 Specification
3. Virtex-6 Family Overview ([DS150](#))
4. Spartan-6 Family Overview ([DS160](#))
5. 7 Series FPGAs Overview ([DS180](#))
6. LogiCORE AXI Interconnect IP Data Sheet ([DS768](#))
7. Vivado documentation [website](#)

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Revision History

The following table shows the revision history for this document:

Date	Version	Description of Revisions
3/1/11	1.0	Initial Xilinx Release
6/22/11	2.0	Updated to ISE 13.2. Updated for Artix-7, Virtex-7, and Kintex-7.
7/25/12	2.1	Updated to Vivado 2012.2 and AXI Strobe support for single AXI transfers.

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